```
                continue

        self._append_master_excel_rows(grouped_rows)

    def _append_master_excel_rows(self, grouped_rows):
        if not self.icao_master_enabled or not grouped_rows:
            return
        if self._master_wb is None:
            self._load_or_create_master_wb()
        if self._master_wb is None or self.icao_master_path is None:
            return

        try:
            index_ws = self._master_wb["ICAO_INDEX"]
        except KeyError:
            index_ws = self._master_wb.create_sheet("ICAO_INDEX", 0)
            index_ws.append(["ICAO", "SheetName", "RecordCount"])

        index_row_for_icao = {}
        for row_idx in range(2, index_ws.max_row + 1):
            key = str(index_ws.cell(row=row_idx, column=1).value or "").strip().upper()
            if key:
                index_row_for_icao[key] = row_idx

        for icao, icao_rows in grouped_rows.items():
            sheet_name = icao[:31]
            ws = self._master_wb[sheet_name] if sheet_name in self._master_wb.sheetnames else self._master_wb.create_sheet
    | (sheet_name)
            self._ensure_excel_headers(ws, self._master_headers)
            for record in icao_rows:
                ws.append(
                    [
                        record["timestamp"],
                        record["icao"],
                        record["callsign"],
                        record["latitude"],
                        record["longitude"],
                        record["altitude"],
                        record["heading"],
                        record["speed"],
                        record["vertical_rate"],
                    ]
                )
                ws.cell(row=ws.max_row, column=4).number_format = "0.00000000"
                ws.cell(row=ws.max_row, column=5).number_format = "0.00000000"

            current_count = max(0, ws.max_row - 1)
            if icao in index_row_for_icao:
                r = index_row_for_icao[icao]
                index_ws.cell(row=r, column=2, value=sheet_name)
                index_ws.cell(row=r, column=3, value=current_count)
            else:
                index_ws.append([icao, sheet_name, current_count])
                index_row_for_icao[icao] = index_ws.max_row

        try:
            self._save_workbook_path(self._master_wb, self.icao_master_path)
        except Exception:
            pass

    def _ensure_excel_headers(self, ws, headers):
        current = [cell.value for cell in ws[1]] if ws.max_row >= 1 else []
        if current[: len(headers)] == headers:
            return
        if ws.max_row == 1 and not any(current):
            for idx, value in enumerate(headers, start=1):
                ws.cell(row=1, column=idx, value=value)
            return
        looks_like_header = any(
            str(v).strip().lower() in {"timestamp", "icaoaddress", "icao"}
            for v in current
            if v is not None
        )
```